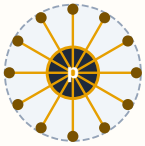
a Stage 0: 3D dark-core prior & auto-crop



6 diameters × 4 radii
$r \in [1.35, 3.30]$ mm

Two-sided radial contrast $C_d(\mathbf{p}, r)$

$\min(I(\mathbf{p} + r\mathbf{u}_d), I(\mathbf{p} - r\mathbf{u}_d)) - I(\mathbf{p})$

Rank-2 / Rank-3 order statistic $S_0(\mathbf{p})$

Tolerates 1 touching neighbour

Auto template crop (k_lim_auto): 0.27 s Rust CPU

b Stage 1: Native-grid 3D ResUNet & SSL

6-Ch Input X

1. T2W I_{norm}
2–3. S_0, S_{agg}
4–6. D_{wall}, G_z, M

→

Anisotropic 3D ResUNet

1×3×3 stem

SE-Res blocks

→

4 Output Heads

- P_{mask} (shaft)
- P_{peak} **(2 peaks)**
- (V_x, V_y) tilt
- P_{obt} (obturator)

51-vol. pelvic SSL pre-training (vessel negatives)

+ Multi-rater consensus weighting (40 dual-traced)

c Stage 2A/B: Triplet MPNN & bundle GraphTF

Stage 2A: 4× Triplet MPNN

Scores slice-to-slice bending
$\cos \angle(\mathbf{v}_{ij}, \mathbf{v}_{jk})$ across $z-1, z, z+1$

+ Collinear gap stitching

Bridges ≤ 20 mm dropouts

→

Stage 2B: Bundle GraphTF

26-D node · 8-D pair edge

2× local + 1× global attn.

+ Implant-region XY gate

Keep if calibrated $p_m \geq \tau$

Pre-trained on 1,500 SynthBrachy3T bundles (78.2%)

Reason-coded filter (kept / low-p / dup.) + Poly3

d Stage 2C: 1D tip localizer & conformal sphere

1D Tip Profile

91 steps (0.5 mm)
$s \in [-25, +20]$ mm

22 channels

(24 ch in 4d)

→

1D Tip Localizer

3× Dilated ResNet

($d = 1, 2, 4$)

+ 4-Head Attn.

Stopping prob. q_t

→

3D Tip Sphere



$r_{90} = q_{90}\,\hat{\sigma}_{\text{tip}}$

Split-conformal 3D tip sphere: 90.2% empirical cov.

31.5% narrower median radius than unscaled intervals